

Antiprotozoal Agents

Protozoal Infections

Parasitic protozoa: live in or on humans

- Malaria
- Leishmaniasis
- Amebiasis
- Giardiasis
- Trichomoniasis
- Pneumocystis
- African trypanosomiasis (sleeping disease)

Transmission

- ✓ Person to person
- ✓ Ingestion of contaminated water or food
- ✓ Direct contact with the parasite
- ✓ Insect bite (mosquito)

Malaria

- **Malaria** is a mosquito-borne infectious disease of humans and other animals caused by parasitic protozoa (a group of single-celled microorganisms) belonging to the genus *Plasmodium*
- Transmitted by the infected female anopheles mosquito
- The protozoan organisms causing malaria are the genus *plasmodium*
- Only four of approximately 100 spp cause malaria in humans. The remaining affect birds, monkeys, livestock, rodents and reptiles
- The four spp that affect human are *P. falciparum*, *P. vivax*, *P. malariae*, and *P. ovale*
- Malaria is present all over the tropics and is now extended to more than 40% of the world's population

Malarial Parasite

Two interdependent life cycles

- Sexual cycle: occurs in the mosquito
- Asexual cycle: occurs in the human
 - Knowledge of the life cycles is essential in understanding antimalarial drug treatment
 - Drugs are effective only during the **asexual** cycle

Plasmodium Life Cycle

Asexual cycle: two phases

- **Exoerythrocytic phase**
 - Occurs “outside” the erythrocyte
 - Also known as the **tissue phase**
- **Erythrocytic phase**
 - Occurs “inside” the erythrocyte
 - Also known as the **blood phase**

Erythrocytes = RBCs

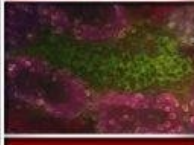
Stage1 - Transmission to human

Stage 1



Transmission to human

Stage 2



Liver invasion

Stage 3



Red blood cell stages

Stage 4



Transmission to mosquito

Stage 5



Mosquito stages

Malaria lifecycle

Stage 1 Transmission to human

Sporozoites
Mosquito injects malaria parasite (sporozoites) into human.

Stage 2 Liver stage

Sporozoite
Liver cell
10 days
Merozoites
Sporozoites invade liver cells. After 10 days thousands of merozoites emerge from the liver cells into the blood.

Stage 3 Red blood cell stage

Merozoites invade red blood cells. After 48 hours, up to 20 daughter merozoites are released from each red blood cell into the blood.

Merozoite
Red blood cell
48 hours
Daughter merozoites
10 days
Gametocytes
Sexual forms of the parasite, called gametocytes, develop over 10 days in some red blood cells.

Stage 4 Transmission to mosquito

Gametocytes
Gametocytes taken up from human by mosquito.

Stage 5 Mosquito stage

Sporozoites
Oocyst
Ookinete forms an oocyst containing thousands of sporozoites. Oocyst bursts releasing sporozoites that migrate to the salivary gland.
Ookinete
Zygote
Gametes
Gametocytes
Gametocytes mature into gametes. Sexual reproduction results in the formation of an ookinete.

This lifecycle is representative of *Plasmodium falciparum*. Numbers may vary for other *Plasmodium* species.

Life cycle of malarial parasites

- Bite of an infected female *Anopheles* mosquito carrying malaria-causing parasites called **sporozoites** and injects them in to human skin dermis.
- After inoculation, the sporozoites travel to the liver and invade liver cells with some getting in to lymph nodes through lymph vessel.
- the sporozoites grow, divide, and produce tens of thousands of haploid forms, called **merozoites**; per liver cell. Then the merozoites rupture the liver cells and re-enter the bloodstream.

- Beginning a cycle of invasion of red blood cells, asexual replication, and release of newly formed merozoites from the red blood cells repeatedly over 1-3 days.
- Some of the merozoite-infected blood cells leave the asexual cycle and instead develop into sexual forms of the parasite, called male and female **gametocytes** that circulate in the bloodstream.
- When a mosquito bites an infected human, it ingests the gametocytes.

- In the mosquito gut; the gametocytes, which develop further into mature sex cells called gametes (Male and female) fuse to form diploid zygotes.
- which develop into actively moving ookinetes that burrow into the mosquito midgut wall and form oocysts. Each oocyst grows, produces thousands of active haploid forms called **sporozoites**.
- The cycle of human infection re-starts when the mosquito takes a blood meal, injecting the sporozoites from its salivary glands into the human bloodstream
- Some of the released merozoites are sequestered in vital organs (brain, and heart) where they continue to grow.

Types of malaria

Plasmodium falciparum

- This spps is estimated to cause approximately 50% of all malaria
- *P. falciparum malaria is the most serious* and most lethal form of the disease
- It is responsible for 90% of the deaths from malaria
- The parasitemia achieved can be quite high and will be associated with an increased incidence of serious complications
- *P. falciparum* leaves the patient so weak because it infects up to 65% of the patients erythrocytes
- It has an incubation period (time from mosquito bite to clinical symptoms) of 1-3 weeks (average of 12days)

- The symptoms of *P. falciparum* malaria are periodic acute attacks of chills and fever, profuse sweating, enlarged spleen and liver, vomiting, anemia, abdominal pain, headaches, and lethargy

Plasmodium vivax

- It is the second most common spp accounting for about 40% of all malaria cases. It can be very chronic, because it can reinfect liver cells
- It has an incubation period of 1-4 weeks (average of 2 weeks)
- Can cause spleen rupture and anemia
- It is characterized by periodic acute attacks of chills and fever, profuse sweating, enlarged spleen and liver, anemia, vomiting abdominal pain, headaches, and lethargy

NB:-If *P. vivax* malaria is not treated, the symptoms may subside for several weeks or months and then recur

Plasmodium malariae

- It accounts for 9.8% of all malaria cases
- It has an incubation period of 2-4 weeks (average of 3 weeks)
- The asexual cycle occurs every 72 hours
- Mildest form of malaria and non-relapsing
- Unlikely to be resistant

Plasmodium Ovale

- This is the least common type of *Plasmodium*
- Has an incubation period of 9-18 days (14 days average)
- Relapse has been known to occur.

Symptoms of **Malaria**

Central

- Headache

Systemic

- Fever

Muscular

- Fatigue
- Pain

Back

- Pain

Skin

- Chills
- Sweating

Respiratory

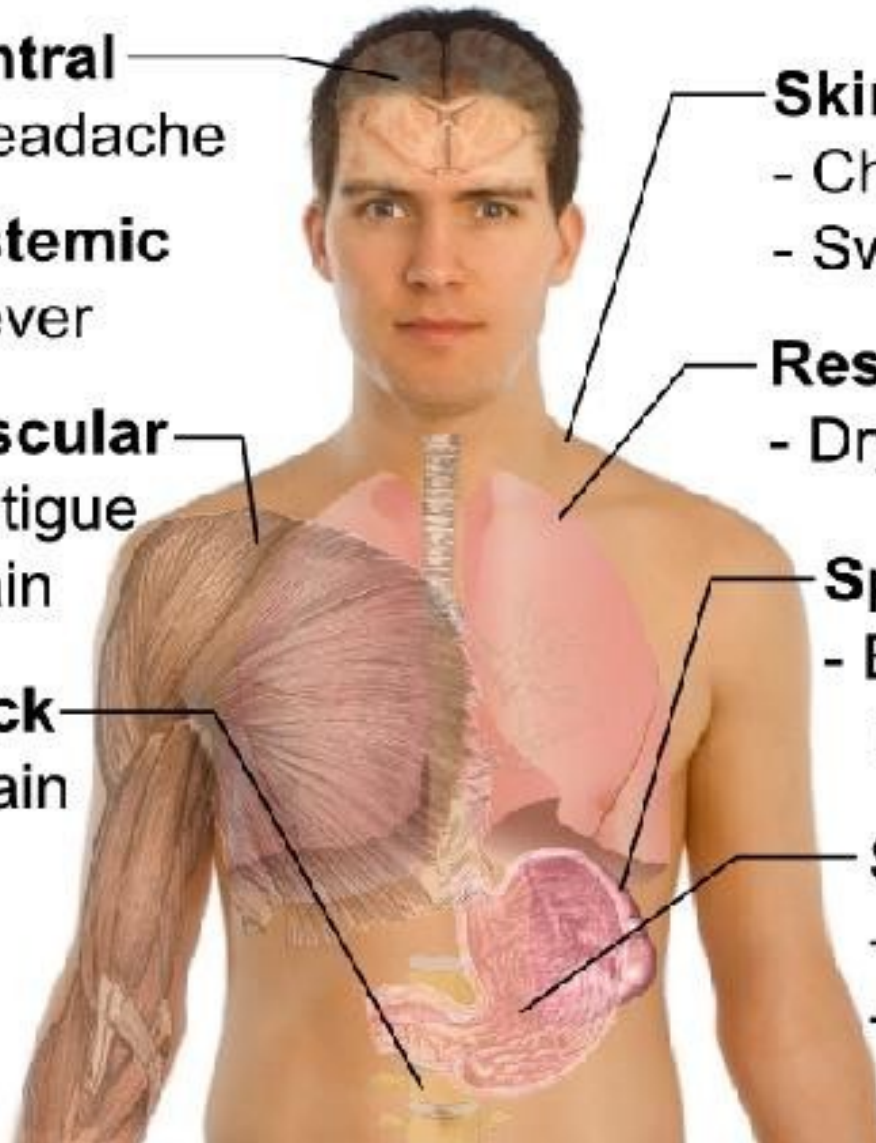
- Dry cough

Spleen

- Enlarge-
ment

Stomach

- Nausea
- Vomiting



Antimalarials

The main classes of active antimalarials are:-

A. Quinoline derivatives

- 8-Aminoquinolines
- 4-Aminoquinolines
- Aryl-alcohols
- 4-methanolquinolines

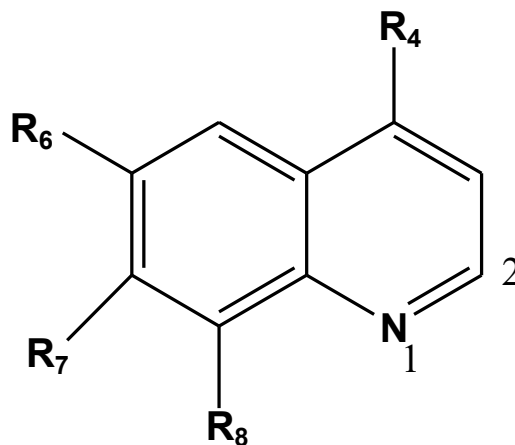
B. Hydroxynaphtoquinone derivative

C. Folate antagonists

D. Antibiotics/Antimicrobials

E. Newer Agents

- Artemisinin derivatives
- Mefloquine



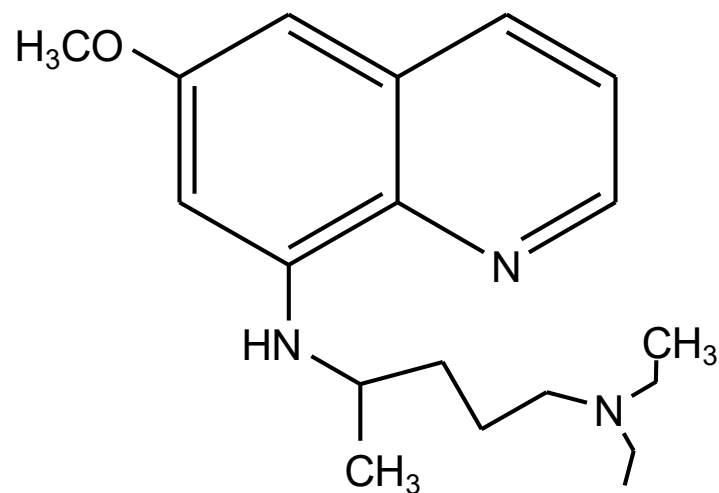
Quinoline

- The amino quinoline nucleus is very important for binding
- Substitution on the C3 position of the ring produces inactive compound
- Alkyl amino/diamino group at C4 or C8 position is very important
- Optimum activity if side chain = 2-6 carbons, if more or less activity will decrease
- Terminal amino group may be 1^o, 2^o or 3^o for 4- amino quinolines, 3^o for others
- C6-Ome is important however, C6-OEt, C6 halogens replacements showed reduced activity
- Substitution of C7 with halogen increases activity but other groups decrease activity
- Methylation at position C8 position decrease activity

Antimalarials (quinolines)

8-Aminoquinolines

- This is the only class of **gametocytocides**
- Pamaquine (prevent relapses) was first introduced for treatment of malaria in 1926 and it has been replaced with primaquine



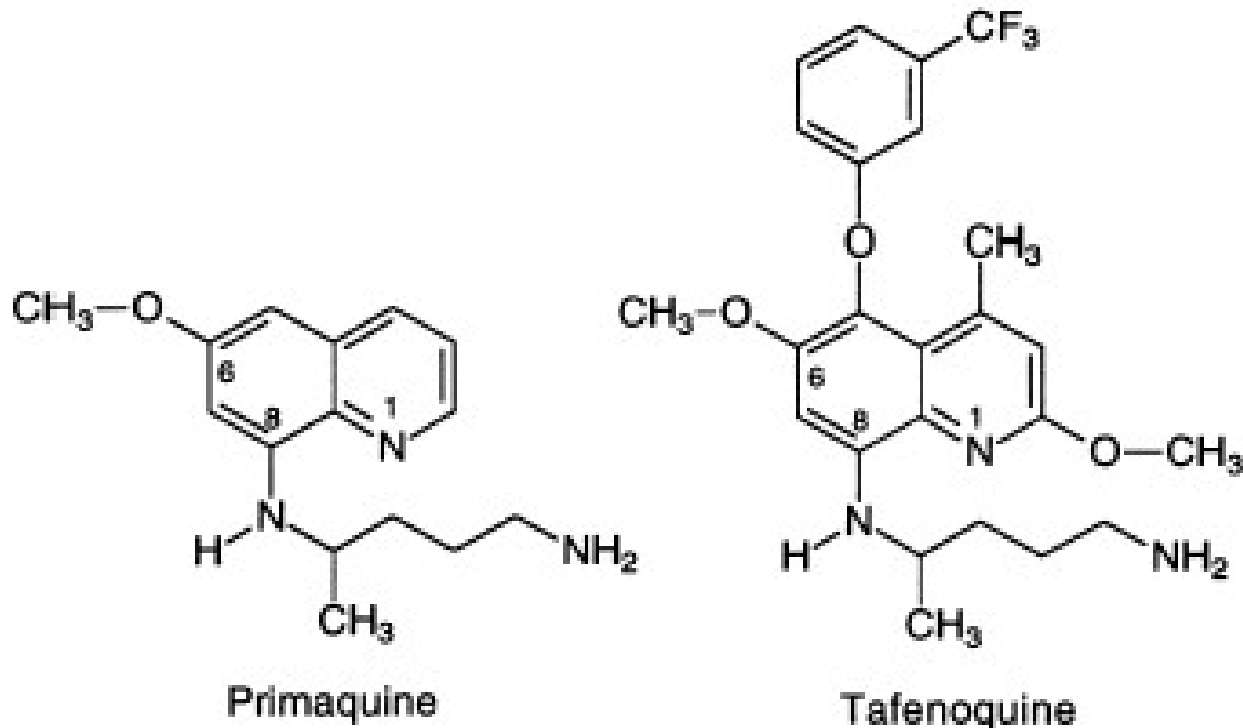
Pamaquine H_3C

Antimalarials (quinolines)

- Primaquine is the only agent that can lead to radical cures of the relapsing forms of *P. vivax* and *P. oval*
- However, primaquine was recently reconsidered for malaria chemoprophylaxis to eliminate *P. falciparum* at the early stage of infection, when parasite develops in the liver, thus preventing the clinical disease
- Despite its good oral absorption, this molecule has a short half-life and needs to be administered daily.

Antimalarials (quinolines)

- Tafenoquine is a primaquine analog with a longer elimination half-life (14 days compared to 4 hours for primaquine)
- It has also a larger therapeutic index than primaquine. This molecule may be useful for chemoprophylaxis of *P. falciparum* and for prevention of relapses of *vivax* malaria



Antimalarials (quinolines)

4-Aminoquinolines

- The main antimalarials are the 4-aminoquinolines because they have proven to be **the most highly successful class of compounds for the treatment and prophylaxis of malaria**
- They are **easily synthesized, cheap, and generally well tolerated**
- These compounds, as well as the quinoline-alcohols, are active against the intra-erythrocytic stages of the parasite

Antimalarials (quinolines)

- The 4-aminoquinolines are able to accumulate to high concentrations within the acid food vacuole of *Plasmodium*, to kill the parasite

Chloroquine (CQ)

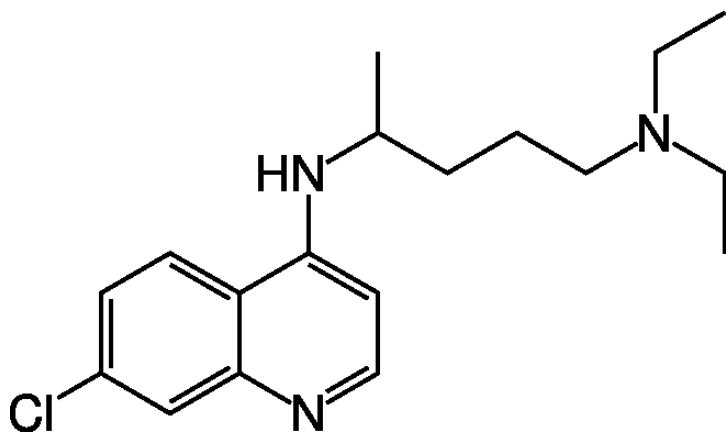
- Was introduced in 1944 –1945 and soon became the mainstay of therapy and prevention, since this drug was **cheap, non-toxic, and active against all strains of malaria parasites**
- In 1994, CQ was the third most widely consumed drug in the world after aspirin and paracetamol
- CQ is not active against the pre-erythrocytic plasmodium and does not cause radical cure

Antimalarials (quinolines)

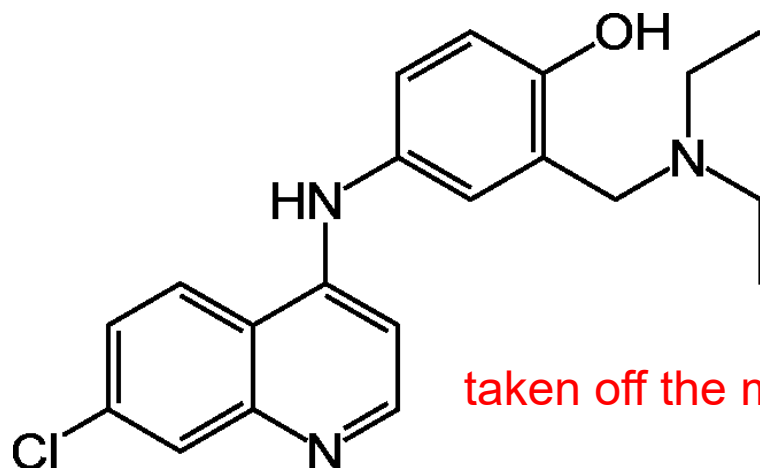
- Chloroquine has activity against the blood stages of *Plasmodium ovale*, *P. malariae*, and susceptible strains of *P. vivax* and *P. falciparum*.
- Widespread resistance in most malaria-endemic countries has led to decline in its use for the treatment of *P. falciparum*, although it remains effective for treatment of *P. ovale*, *P. malariae*, and, in most regions, *P. vivax*.
- Used to treat nonfalciparum and sensitive falciparum malaria.
- Primaquine must be added for the radical cure of *P. vivax* and *P. ovale*, because of these species does not eliminate

Antimalarials (quinolines)

- Rapidly and almost completely absorbed from the gastrointestinal tract.
- Very large apparent volume of distribution of 100-1000 L/kg



Chloroquine



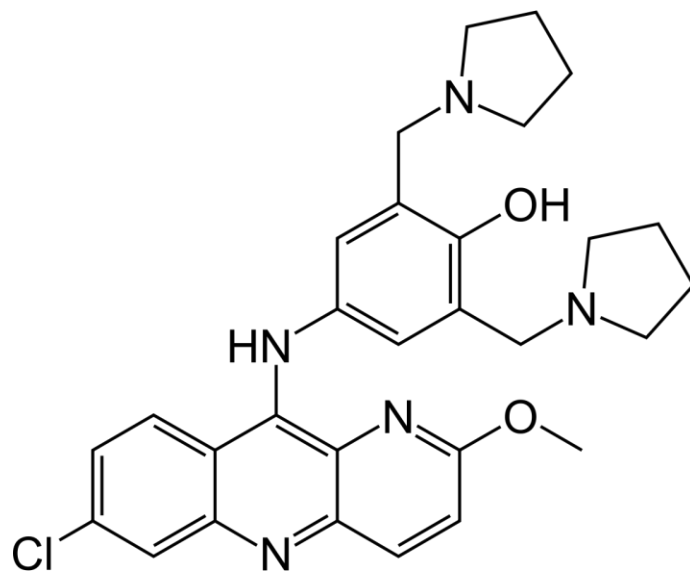
taken off the market

Amodiaquine

Antimalarials (quinolines)

Pyronaridine

- Is a synthetic drug widely used in China that may have utility for multiresistant *falciparum* malaria
- It seems likely that drug resistance would emerge rapidly if pyronaridine is used as monotherapy



Antimalarials (quinolines)

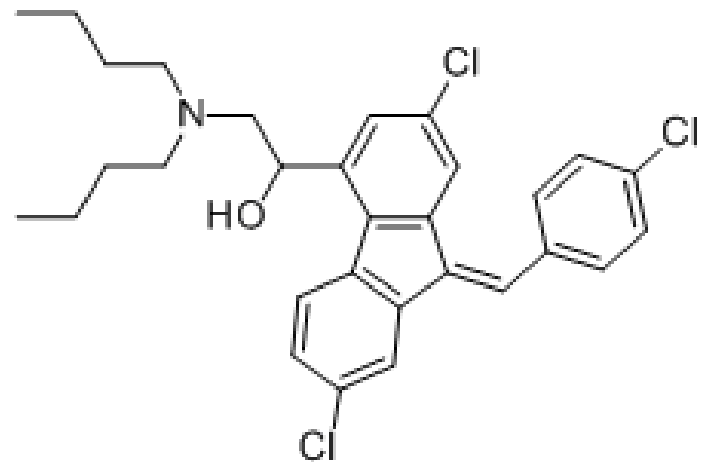
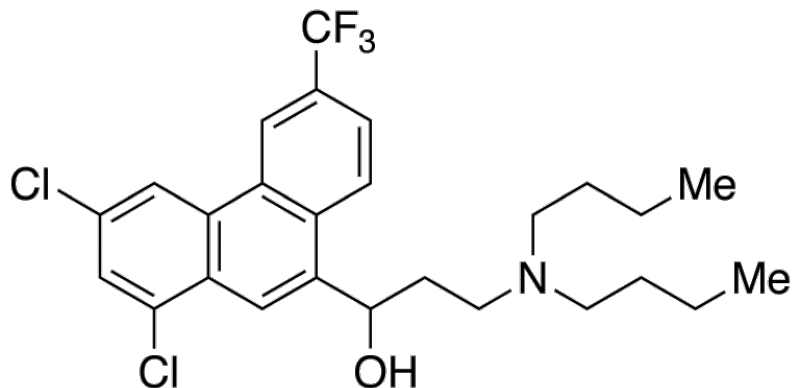
Aryl-alcohols

Halofantrine

- Is effective against chloroquine-resistant malaria
- Despite this, cardiotoxicity has limited its use as a therapeutic agent.

Lumefantrine

- is an antimalarial drug that is only used in combination with artemether (**Coartem**)



Antimalarials (quinolines)

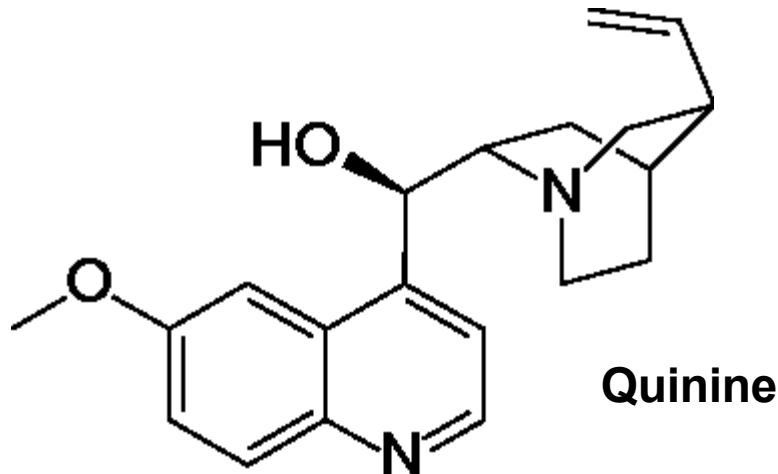
4-methanolquinolines (Cinchona alkaloids)

Quinine

- The active ingredient of cinchona bark, had the longest period of effective use, but there is now a decrease of the clinical response of *P. falciparum* in some areas
- Reserved for severe infestations and for malarial strains that are resistant to other agents, such as chloroquine
- Highly effective blood schizonticide against the four species of human malaria parasites.
- **Clinical Uses:** mainly for chloroquine-resistant falciparum malaria, especially for cerebral malaria.

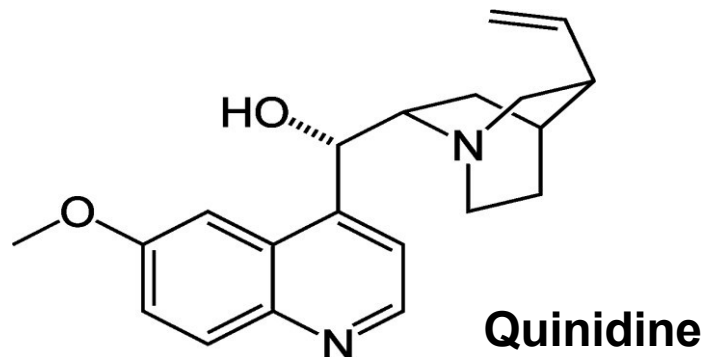
Antimalarials (quinolines)

- The addition of a single dose of artemisinin enhances the parasite elimination rate and thus increases the cure rate
- Association of quinine and clindamycine significantly shorten the duration of treatment with respect to quinine used alone



Antimalarials (quinolines)

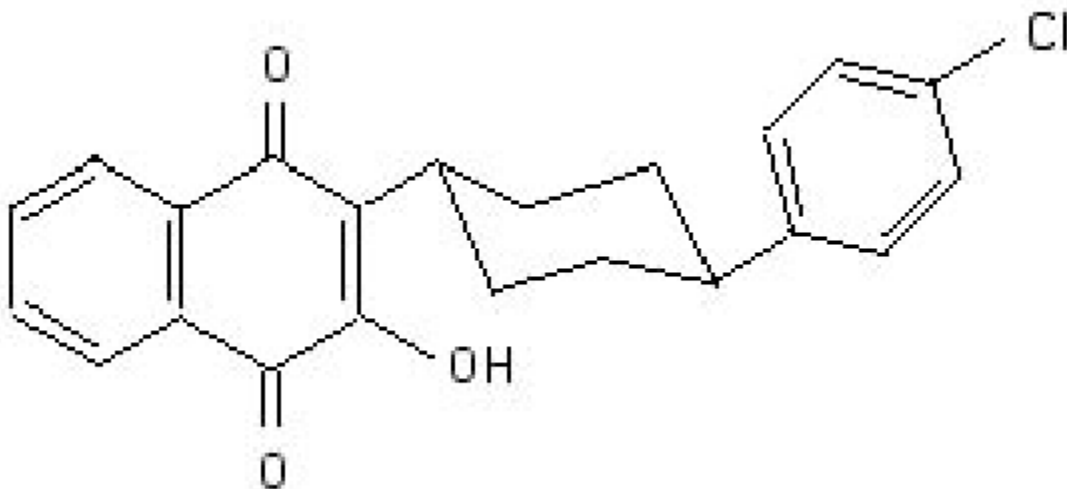
- **Quinidine** is a pharmaceutical agent that acts as a class I antiarrhythmic agent (Ia) in the heart. It is a stereoisomer of quinine, originally derived from the bark of the cinchona tree
- Quinidine is the more cardiotoxic than quinine
- Cardiac monitoring including serial blood pressure measurements should always be in place when quinidine is administered, to monitor for prolongation of the QT interval and ventricular tachycardia.



Hydroxynaphthoquinone derivative

Atovaquone

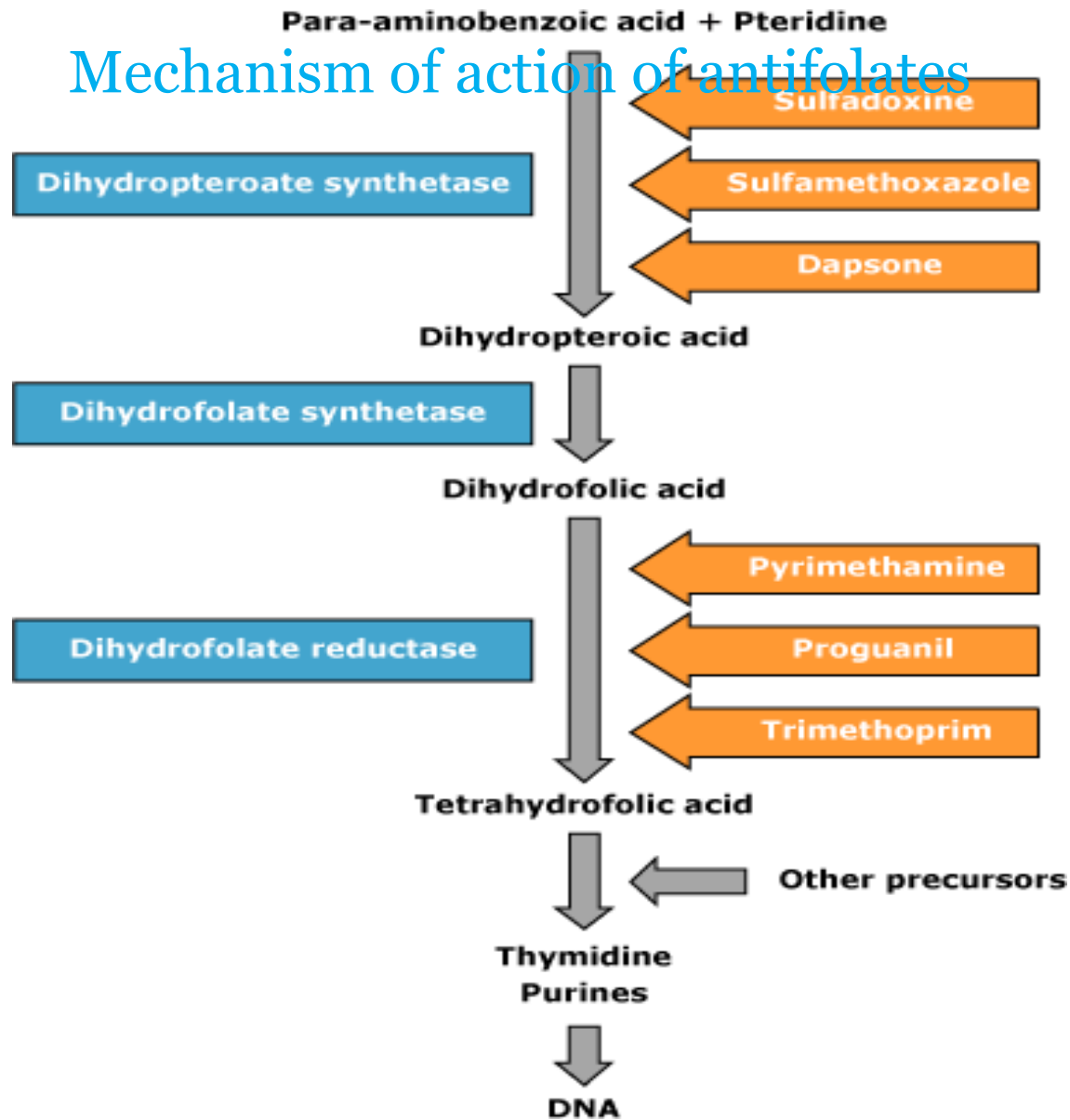
- This drug currently given as combination form with proguanil (malarone)



Folate antagonists/Antifolates

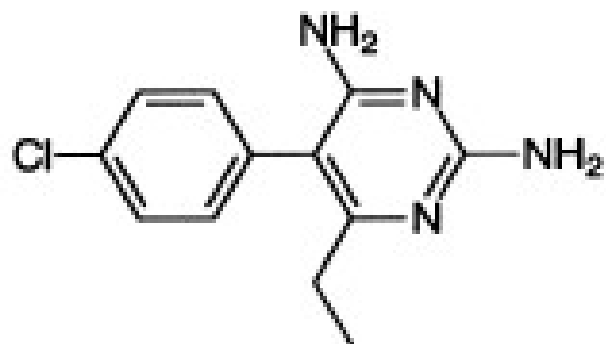
- Antifolates include guanidine analogues (proguanil hydrochloride, cycloguanil, chloroproguanil), pyrimidine analogues (pyrimethamine and trimethoprim) and sulfones (sulfonamides and dapsons).
- These drugs act synergistically to target enzymes involved in folate synthesis, a pathway required for parasite DNA synthesis.
- There are two groups of antifolates:-
 - ✓ The dihydrofolate reductase (DHFR) inhibitors like pyrimethamine and proguanil
 - ✓ The dihydropteroate synthase (DHPS) inhibitors that are sulfones and sulphonamides like sulfadoxine and dapsons, respectively

Mechanism of action of antifolates

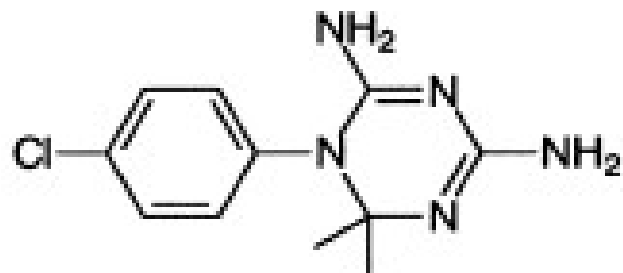


- Due to a marked synergistic effect, a drug of the first group is usually used in combination with a drug of the second one
- Pyrimethamine – Sulfadoxine (SP) is the most widely used combination
- SP is also particularly prone to rapid emergence of resistance
- The development of the combination of proguanil and atovaquone (Malarone) also provide another useful way for the treatment

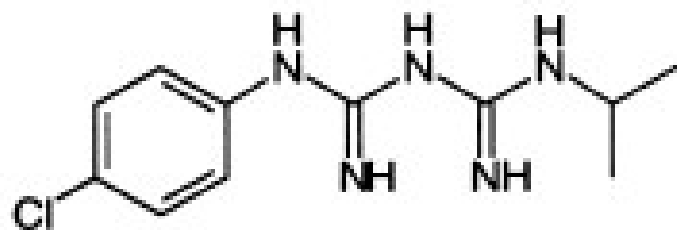
- Current practice in threatening cases of malaria is based on the concept of **combination therapy**, since this offers several advantages
 - ✓ reduced risk of treatment failure
 - ✓ reduced risk of developing resistance
 - ✓ enhanced convenience and
 - ✓ reduced side effects



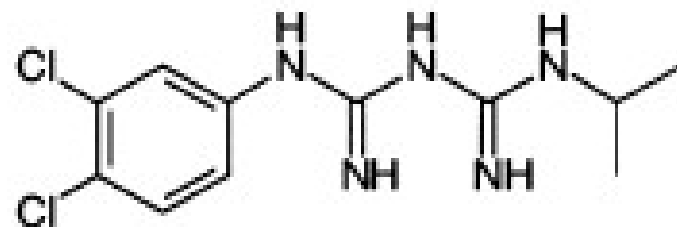
Pyrimethamine



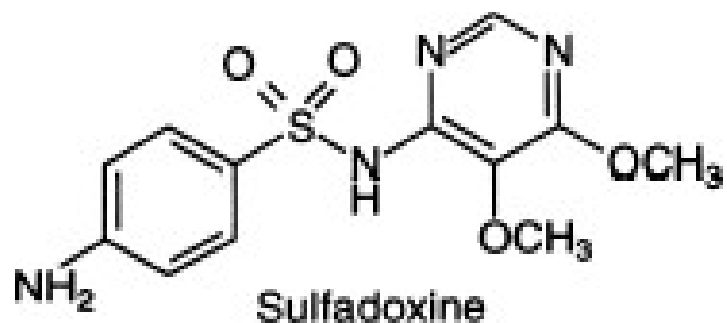
Cycloguanil



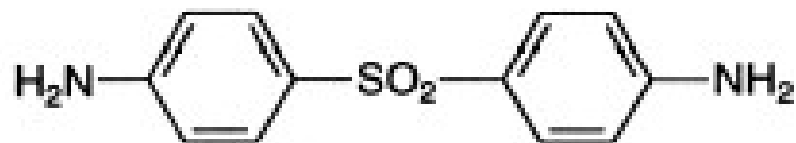
Proguanil



Chlorproguanil



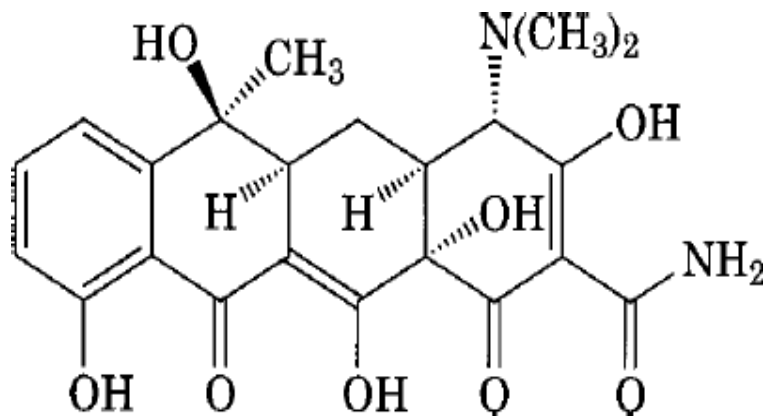
Sulfadoxine



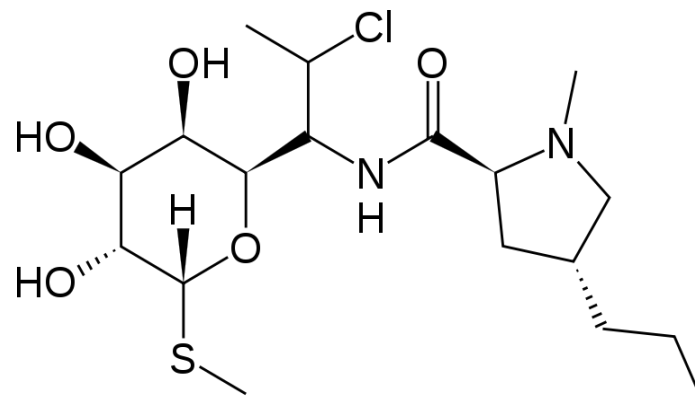
Dapsone

Antibiotics/Antimicrobials

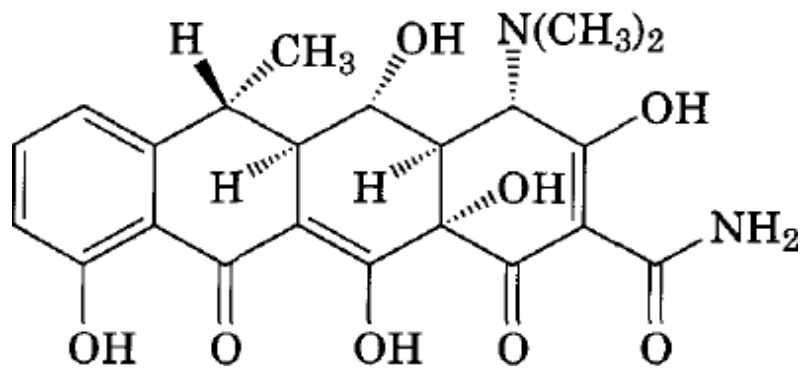
- Tetracycline, doxycycline and clindamycin target prokaryotic protein synthesis
- In malaria parasites, these drugs appear to target the apicoplast, an organelle derived from prokaryotic ancestors
- They have relatively slow antimalarial activity
- They are typically paired with fast-acting antimalarials (usually quinine)
- Doxycycline has a longer half life than tetracycline so is used more commonly
- Resistance has not been detected to tetracycline, doxycycline or clindamycin



Tetracycline



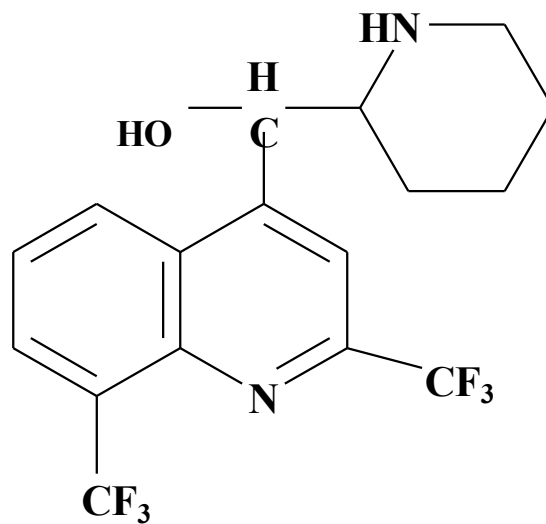
Clindamycin



Doxycycline

Newer Antimalarials

- Mefloquine:**-Is structurally related to quinine, and its long half-life (14 –21 days) has probably contributed to the rapid development of resistance
- For this reason, mefloquine should be used in combination with other antimalarial agents
 - **Clinical Uses:-** Chemoprophylaxis, mainly for chloroquine-resistant falciparum malaria.

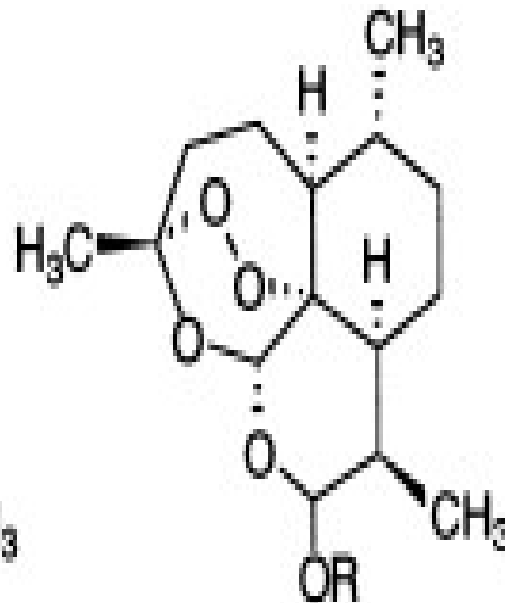
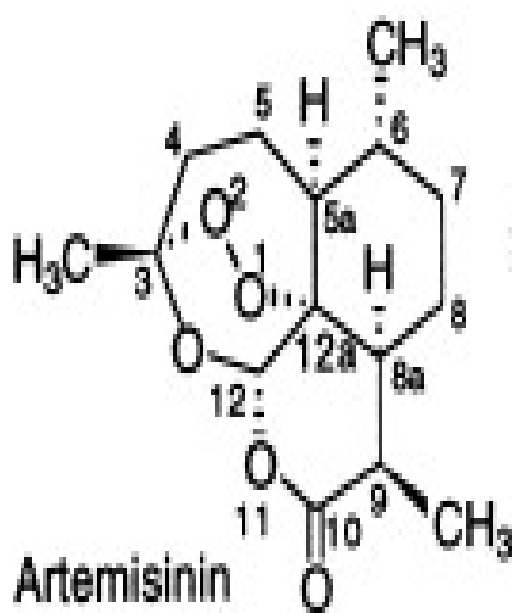


Artemisinin derivatives (newer agents)

- Artemisinin derivatives are the most potent and rapid acting blood schizonticides
- Most of artemisinin derivatives are highly active against *plasmodium* genera that cause malaria.
- Four compounds have been used, the parent one, **artemisinin**, extracted from *Artemisia annua* and four derivatives that are actually more active than artemisinin itself
 - ✓ **dihydroartemisinin**
 - ✓ **artesunate**: a water-soluble hemisuccinate
 - ✓ **artemether** :oil-soluble
 - ✓ **artemotil (arteether)** :oil-soluble

- All of them are readily metabolized to the biologically active metabolite, dihydroartemisinin
- Artemisinin is active at nanomolar concentrations *in vitro* on both CQ-sensitive or -resistant *P. falciparum* strains
- These classes of drugs are also less toxic than other antimalarial agents.
- Artemisinin and its derivatives appear to be the best alternative for the treatment of severe malaria

- Institute of research has patented a stable, water-soluble derivative called artelinic acid that is now being tested in animals.



R = H, Dihydroartemisinin

R = Me, Artemether

R = Et, Arteether

R = $\text{OCO}(\text{CH}_2)_2\text{COONa}$

Artesunate

R = $\text{OCH}_2(p\text{-C}_6\text{H}_4)\text{COOH}$

Artelinic acid

- A key advantage of these endoperoxide-containing antimalarial agents, which have been used for nearly two decades, is the absence of any drug resistance
- The major drawback of artemisinin derivatives is their short half-life. When used in monotherapy, a treatment as long as 5 days is required for complete elimination of the parasites
- They are then preferentially used in combination with other antimalarial agents such as sulfadoxine-pyrimethamine, benflumetol, or mefloquine to increase cure rates and to shorten the duration of therapy in order to minimize the emergence of resistant parasites

Mechanism of action

- These compounds have presence of endoperoxide bridge
- Endoperoxide bridge interacts with heme in parasite
- Heme iron cleaves this endoperoxide bridge
- There is generation of highly reactive free radicals which damage parasite membrane by covalently binding to membrane proteins

Common combinations

- Chloroquine/sulfadoxine-pyrimethamine
- Quinine/ Doxycycline
- Artemether/lumefantrine (**Coartem/Banflumetol**)
- Atovaquone-proguanil (**Malarone**)
- Artesunate/amodiaquine (**Coarsucam**)
- Sulfadoxine-pyrimethamine (**Fansidar**)
- Artesunate + sulfadoxine-pyrimethamine (**Supysunate**)
- Artesunate + mefloquine (**Artequin**)
- Artemether + lumefantrine (Banflumetol) is the most viable artemisinin combination treatment available at the moment

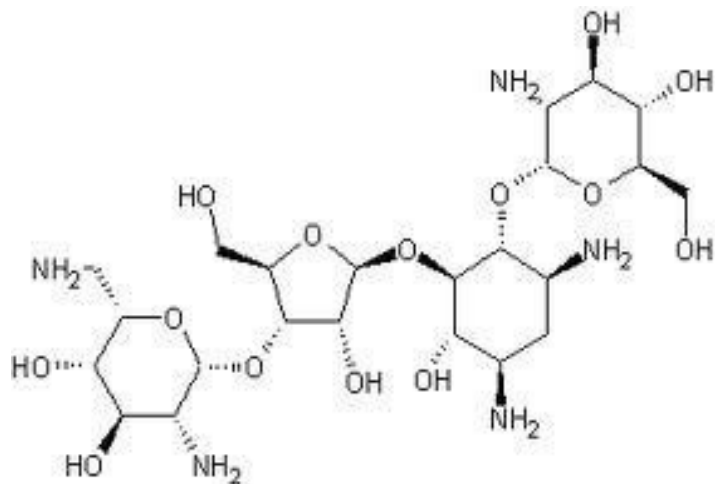
WHO Recommendations

- ✚ WHO recommends that all countries experiencing resistance to conventional monotherapies should use combination therapy, preferably containing artemisinins [ACTs -artemisinin-based combination therapies].
- ✚ WHO recommends the following therapeutic options:-
 1. Artemether/lumefantrine (coartem)
 2. Artesunate + amodiaquine
 3. Artesunate + SP
 4. Artesunate + mefloquine [area with low to moderate transmission.
 5. Amodiaquine+SP

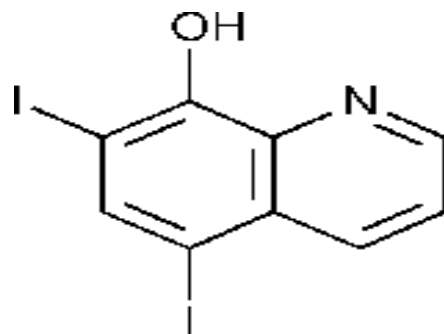
- The advantages of artemisinin based combination therapy (ACT) relate to the unique properties and mode of action of the artemisinin component, which include the following
 - ✓ rapid substantial reduction of the parasite biomass
 - ✓ rapid resolution of clinical symptoms
 - ✓ effective action against multidrug-resistant *P. Falciparum*
 - ✓ reduction of gametocyte carriage, which may reduce transmission of resistant alleles (in areas with low or moderate malaria transmission)
 - ✓ no parasite resistance documented as yet with the use of artemisinin and its derivatives
 - ✓ few reported adverse clinical effects, however pre-clinical toxicology data on artemisinin derivatives are limited.

Amebiasis

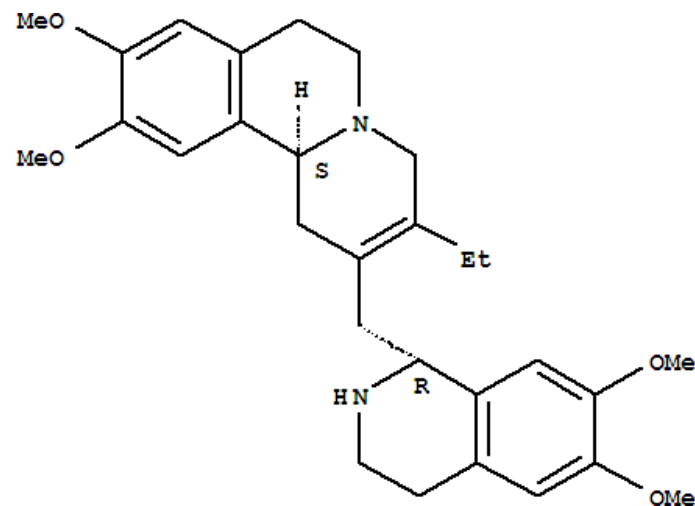
- Amebiasis is an infection of the intestines with a parasite called *Entamoeba histolytica* (*E. histolytica*).
- The parasite is an amoeba, a single-celled organism.
- People can get this parasite by eating or drinking something that's contaminated with it.
- In many cases, the parasite that causes amebiasis lives in a person's large intestine without causing any symptoms. Other times, it causes: diarrhea (which may be bloody), stomach pains, abdominal cramping, loose stool, nausea, loss of appetite and fever
- In rare cases, it can spread into other organs such as the liver, lungs, and brain.



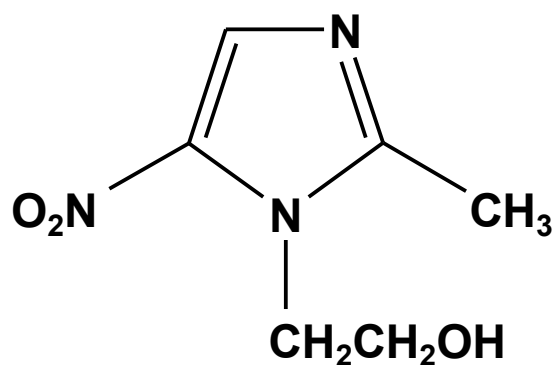
Paromomycin



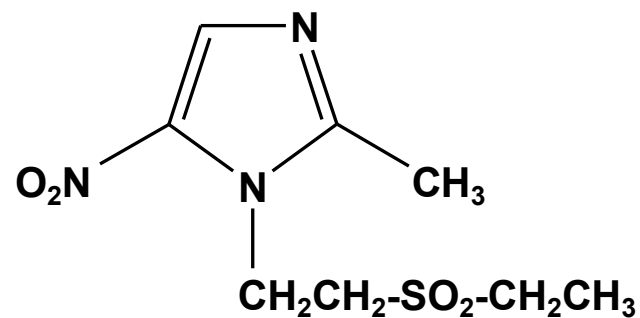
Iodoquinol



Dehydroemetine



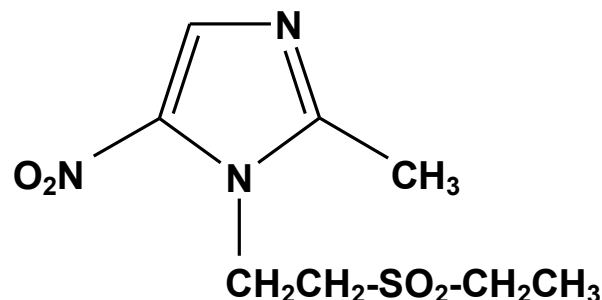
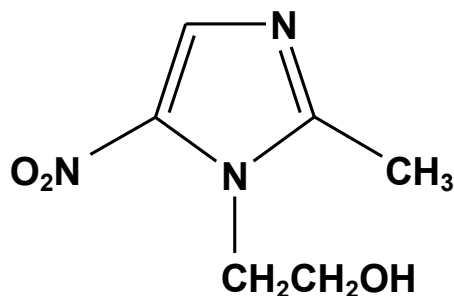
Metronidazole



Tinidazole

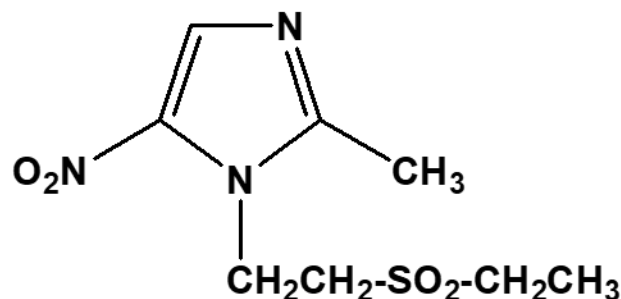
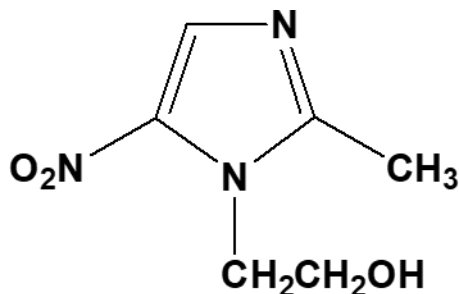
Trichomoniasis

- Trichomoniasis is a common sexually transmitted infection caused by the protozoan *Trichomonas vaginalis*
- Trichomoniasis signs and symptoms for women include: an often foul-smelling vaginal discharge, genital redness, burning and itching, pain with urination or sexual intercourse
- Trichomoniasis rarely causes symptoms in men. When men do have signs and symptoms, however, they might include: irritation inside the penis, burning with urination or after ejaculation, discharge from the penis
- **Treatment:-** Metronidazole or tinidazole



Giardiasis

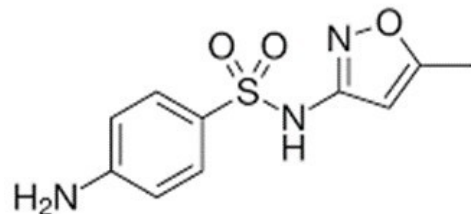
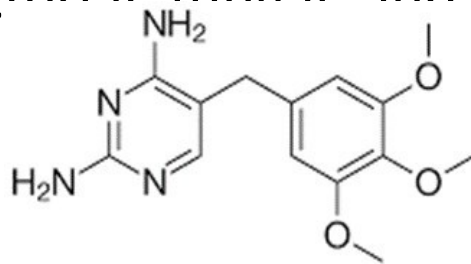
- Giardia infection is an intestinal infection marked by abdominal cramps, bloating, nausea and bouts of watery diarrhea. Giardia infection is caused by a microscopic parasite called *Giardia lamblia* that is found worldwide, especially in areas with poor sanitation and unsafe water.
- *G. lamblia* are found in animal and human feces. These parasites also thrive in contaminated food, water, and soil, and can survive outside a host for long periods of time. Accidentally consuming these parasites can lead to an infection
- **Treatment:**-Metronidazole or tinidazole



Pneumocystis

- *Pneumocystis jiroveci* (previously classified as *Pneumocystis carinii*), the organism that causes pneumocystis pneumonia.
- The symptoms of pneumocystis pneumonia (PCP) include dyspnea (difficulty breathing), nonproductive cough, and fever.
- *Pneumocystis carinii* can also cause extrapulmonary disease in immunocompromised patients (AIDS, etc)

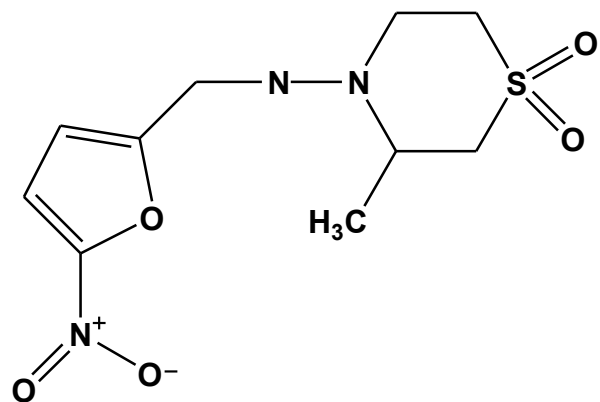
Treatment:-*Co-trimoxazole* (TMP/SMX) is an antibiotic used to treat a variety of bacterial, fungal, and protozoal infections



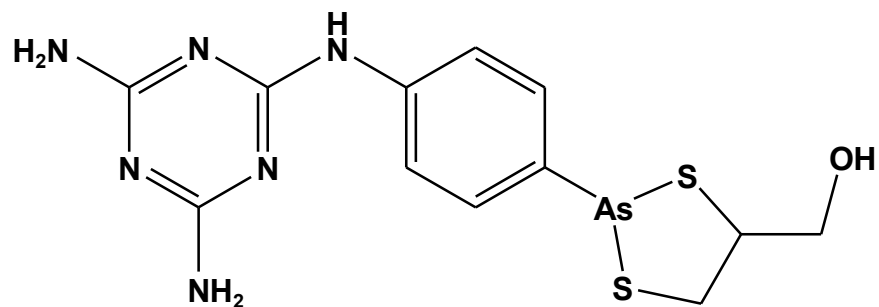
Trypanosomiasis (sleeping sickness)

- It is caused by the hemoflagellates *Trypanosoma brucei rhodesiense* and *Trypanosoma brucei gambiense*.
- There are two types of trypanosomiasis that affect humans, they are divided according to their geographical location
- In Africa the largest number of cases is in Congo.
- Fever, severe headaches, irritability, extreme fatigue, swollen lymph nodes, and aching muscles and joints are common symptoms of sleeping sickness

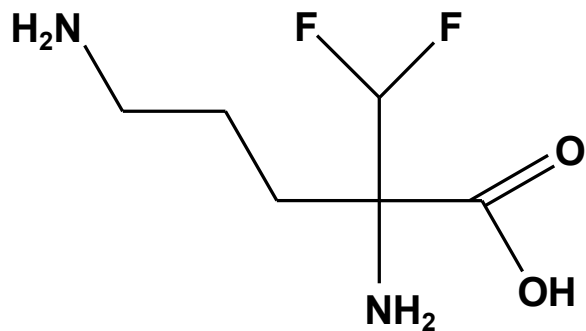
Treatment:- Melarsoprol, Eflornithine, benznidazole or nifurtimox



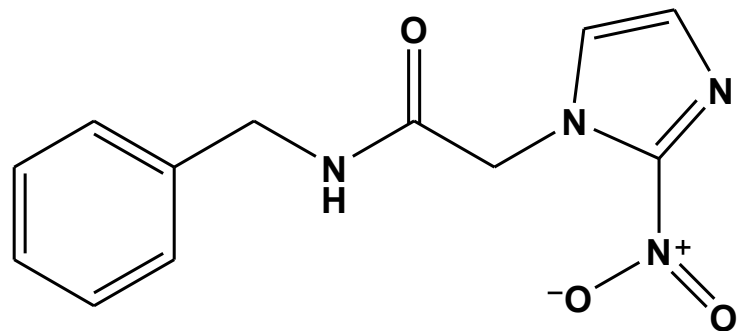
Nifurtimox



Melarsoprol



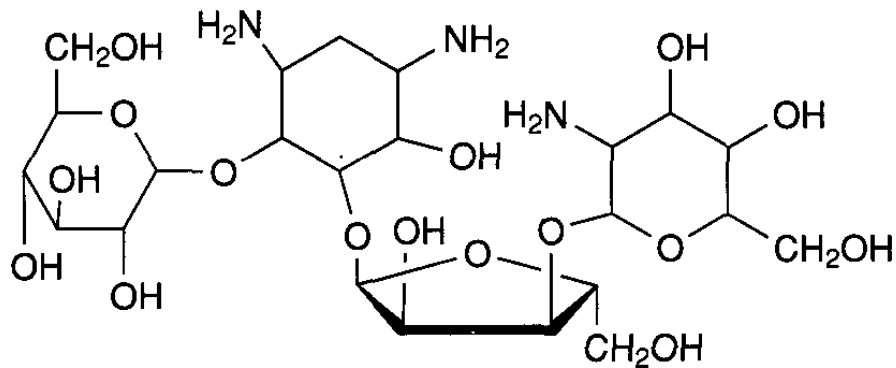
Eflornithine



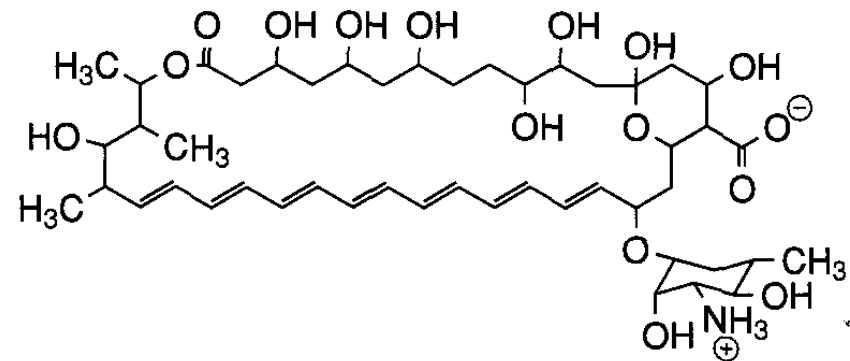
Benznidazole

Leishmaniasis

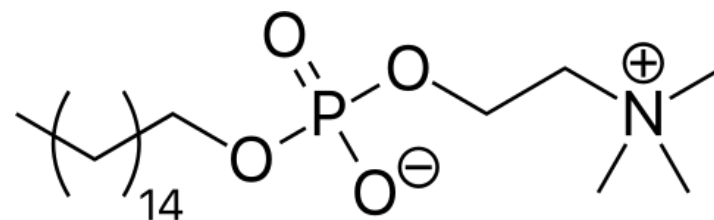
- **Leishmaniasis** is a disease caused by an intracellular protozoan parasite (genus *Leishmania*) transmitted by the bite of a female phlebotomine sandfly.
- *Visceral leishmaniasis (kala azar)* is caused mainly by *Leishmania donovani* in the Indian subcontinent and East Africa
- *Cutaneous leishmaniasis* is caused mainly by *Leishmania tropica*, *L. major*, and *L. donovani*.
- **Treatment**:-Sodium stibogluconate, meglumine antimoniate, amphotericin B, Miltefosine and paromomycin.



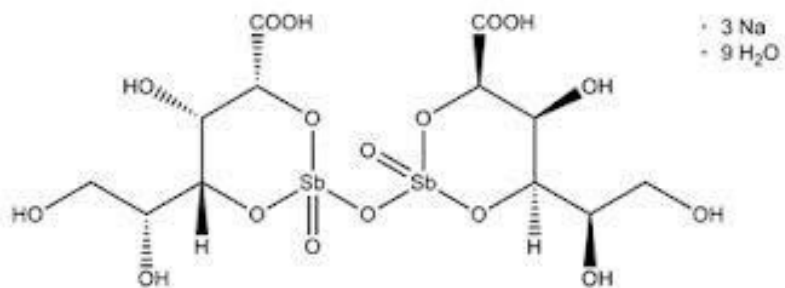
Paromomycin



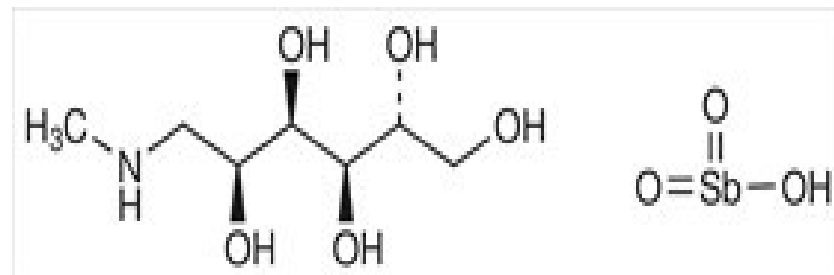
Amphotericin B



Miltefosine



Sodium Stibogluconate



Meglumine antimonate



Visceral leishmaniasis



Cutaneous leishmaniasis